ChatGPT

Related work for “Human + Prompt-Engineered AI vs Human Alone: Evaluating Prompt-Engineered Decision Support in Medicine”

Executive summary

Recent evidence shows that simply giving clinicians access to an LLM does **not** reliably improve diagnostic reasoning, while hybrid human–AI systems can outperform either alone when designed to exploit complementary strengths. Across studies, **prompt design** and **explanation format** emerge as under-controlled levers that can materially affect accuracy, efficiency, and reliance—motivating clinician-in-the-loop trials that explicitly standardize prompts and measure human factors. [1]

Related work text ready to paste

Eric Goh [2] et al. (2024, JAMA Network Open [3]) evaluated whether clinician access to an LLM improves diagnostic reasoning in a single-blind randomized trial (n=50 physicians; 26 attendings/24 residents) using up to six nonpublic clinical vignettes and a structured reflection rubric (top differentials, supporting/opposing features, final diagnosis, next steps). Physicians randomized to conventional resources + LLM (ChatGPT Plus “GPT-4” from OpenAI [4]) did **not** score significantly higher than controls (median reasoning score 76% vs 74%; P=.60), and time per case also did not differ significantly; intriguingly, an LLM-only arm scored higher than the conventional-resources group. Key limitations were the use of a single LLM and the fact that clinicians received no prompt-engineering training and were not required to use the tool consistently—leaving open whether **standardized, prompt-engineered decision support** (rather than ad hoc chatting) yields measurable gains. [5]

Nikolas Zöller [6] et al. (2025, Proceedings of the National Academy of Sciences of the United States of America [7]) studied hybrid “collective intelligence” for open-ended diagnosis at scale by combining 40,762 differential diagnoses from physicians (self-reported tenure: 1,370 attendings, 139 fellows, 2,160 residents) with outputs from five LLMs (Claude 3 Opus from Anthropic [8]; Gemini Pro 1.0 from Google [9]; Llama 2 70B from Meta [10]; Mistral Large from Mistral AI [11]; GPT-4 from OpenAI) across 2,133 text vignettes. They used modular prompt engineering (role/“impersonation,” self-consistency checks, format constraints, and few-shot examples) selected via cross-validation, mapped free-text diagnoses into SNOMED CT, and showed that hybrid human–LLM ensembles outperform single clinicians, clinician-only ensembles, single LLMs, and LLM ensembles—attributing gains to complementary (less correlated) error patterns. However, their evaluation is largely **offline aggregation** rather than a clinician-facing workflow: it does not test how a single clinician’s *interactive* decisions change when presented with prompt-engineered LLM recommendations, nor does it measure time, confidence, or advice uptake—gaps directly aligned with clinician-in-the-loop experiments. [12]

Drew Prinster [13] et al. (2024, Radiology [14]) provide strong evidence that *how* AI is explained can change both performance and trust. In a multicenter prospective randomized user study (n=220 physicians), participants read eight chest radiographs (from MIMIC-CXR) and then received simulated AI advice with either local (feature-based bounding boxes) or global (prototype-based exemplar) explanations; AI correctness and confidence were varied within-participant. Local explanations improved diagnostic accuracy when AI advice was correct and reduced time spent considering advice (greater efficiency), while also increasing a behavioral "simple trust" measure—yet this increased reliance could also be risky when AI advice is incorrect. A key limitation is ecological validity: even with a DICOM viewer, the study used a web interface and simulated advice rather than real-time clinical practice pressures. For LLM-based decision support, this work motivates testing whether prompt-engineered *textual rationales* similarly calibrate (or miscalibrate) reliance—an issue underexplored in diagnosis-focused human–LLM evaluations. [15]

Dhavalkumar Patel [16] et al. (2024, Scientific Reports [17]) directly tested prompt-engineering variants for medical questions by comparing direct prompting, chain-of-thought (CoT), and a modified CoT for GPT-3.5-turbo on 95 USMLE Step 1 questions and 1,000 GPT-4–generated USMLE-style items spanning clinical scenarios and calculations. They found **no significant accuracy differences** across prompt styles (e.g., 61.7% vs 62.8% vs 57.4% on the USMLE sample; P=0.734), suggesting that some commonly recommended prompt patterns may not reliably improve performance in this medical QA setting. Limitations include reliance on exam-style multiple-choice tasks, evaluation of only one model family (GPT-3.5), and rapid model evolution limiting generalizability. The main gap is translation: prompt effects measured on *benchmarks* may not predict the effect of **prompt-engineered outputs used as decision support by clinicians** on complex differential diagnosis tasks. [18]

Finally, Jason Wei [19] et al. (2022, NeurIPS [20]) introduced chain-of-thought prompting as a general mechanism to elicit intermediate reasoning steps via few-shot exemplars, reporting substantial gains on arithmetic, commonsense, and symbolic reasoning—especially for sufficiently large models—and arguing that "chains" can provide an interpretable window into model behavior. The paper also explicitly notes that CoT's human-like reasoning traces do not prove the model is truly reasoning, and that fully characterizing the computations supporting an answer remains open. Although nonclinical, this work underpins the premise that **prompt structure can change reasoning behavior** and can be used to expose rationale-like content; however, it does not test downstream human decision-making, nor does it address whether such rationales calibrate clinician trust or improve diagnostic accuracy in realistic clinical vignettes. [21]

Synthesis across studies. Together, these papers suggest a design tension: (i) prompt engineering and explanation form can change model outputs and user reliance, but (ii) naive deployment of LLM access may not improve clinician reasoning. This motivates controlled experiments that treat prompt design as a first-class intervention and evaluate clinicians' accuracy, efficiency, confidence, and reliance under standardized conditions (Goh et al., 2024; Zöller et al., 2025; Prinster et al., 2024; Patel et al., 2024; Wei et al., 2022). [1]

Research gaps our experiment will fill

- **Prompt-engineering-controlled clinician trials:** Prior clinician trials often provide LLM access without consistent prompt use or training; few isolate *prompt-engineered* decision support as the intervention (Goh et al., 2024). [22]
- **Clinician-facing (not offline) hybridization:** Large-scale hybrid benefits have been shown via offline aggregation, but interactive effects on an individual clinician's diagnosis process (advice uptake, error correction) remain under-tested (Zöller et al., 2025). [23]

- **Human factors beyond accuracy:** Many studies emphasize accuracy/top-k metrics; fewer jointly measure time, confidence, and trust/reliance in a unified design, despite evidence that explanations can shift reliance even when wrong (Prinster et al., 2024). [24]
- **Complex vignette-based differential diagnosis (vs exam QA):** Prompt comparisons on exam-style items may not generalize to ambiguous, multi-differential vignettes and real diagnostic workflows (Patel et al., 2024; Goh et al., 2024). [25]
- **Rationale/explanation format as a safety lever for LLM decision support:** Imaging studies show explanation type can induce overreliance; analogous tests for prompt-engineered *textual rationales* in diagnostic decision support are sparse (Prinster et al., 2024; Wei et al., 2022). [26]

Comparison table of the five papers

Paper (author–year; venue)	Dataset / cases	Participants	AI model type	Prompt engineering (yes/no; type)	XAI / explanations (yes/no)	Main findings	Key lim
Goh et al., 2024; JAMA Netw Open	Up to 6 nonpublic clinical vignettes; structured reflection rubric	50 physicians (attendings + residents)	GPT-4 via ChatGPT Plus	Partial: clinicians used LLM ad hoc; LLM-only arm used iteratively developed 0-shot prompt	No	LLM access did not significantly improve physician diagnostic reasoning or time; LLM-only arm scored higher than conventional-only	Sin sm set cli no in pro an rec use co
Zöller et al., 2025; PNAS	2,133 text vignettes; 40,762 physician differentials; SNOMED mapping	Physicians (counts by tenure reported) + (SI: medical students)	Multiple LLMs (Claude, GPT-4, Gemini, Llama 2, Mistral)	Yes: modular prompt blocks; cross-validated prompt selection	No	Hybrid human–LLM collectives outperform humans-only, LLM-only, and ensembles; gains attributed to complementary errors	Off age lim cli wo me (tin cor adv up

Paper (author–year; venue)	Dataset / cases	Participants	AI model type	Prompt engineering (yes/no; type)	XAI / explanations (yes/no)	Main findings	Ke lim
Prinster et al., 2024; Radiology	8 chest radiographs from MIMIC-CXR; AI advice correct in 6 / incorrect in 2	220 physicians (radiologists + nonradiologists)	Simulated DL-based chest radiograph AI	No	Yes: local vs global explanations; AI confidence varied	Explanation type affects accuracy, efficiency, and behavioral trust; local explanations can increase reliance	Sin int lim cou mo tas
Patel et al., 2024; Scientific Reports	95 USMLE Step 1 + 1,000 GPT-4–generated USMLE-style questions	None (model-only evaluation)	GPT-3.5-turbo	Yes: direct vs CoT vs modified CoT	No	No significant accuracy differences across prompt styles	Exa QA mo pro tec inc rap evo
Wei et al., 2022; NeurIPS	Reasoning benchmarks (arithmetic/ commonsense/ symbolic)	None (model-only evaluation)	Large LMs incl. PaLM 540B	Yes: few-shot chain-of-thought exemplars	Indirect: textual rationale (“chain”) but not validated as faithful	CoT prompting improves reasoning on multiple benchmarks; chains offer an interpretable window	No do tes de ma rea tra fai rer op

Mermaid flowchart of how our experiment extends prior work

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flowchart TD
  subgraph Prior_Work["Prior work (what exists)"]
    A1["Ad hoc clinician access to GPT-4; mixed/no gains (Goh 2024)"]
    A2["Offline hybrid collectives outperform (Zöller 2025)"]
    A3["Explanation type shifts trust/accuracy; possible overreliance (Prinster 2024)"]
    A4["Prompt variants may or may not change medical QA accuracy (Patel 2024)"]
    A5["Prompt structure can elicit reasoning traces (Wei 2022)"]
  end
```

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  subgraph Our_Study["Our extension (clinician-in-the-loop prompt-engineered CDSS)"]
    B1["Participants: clinicians or senior medical students"]
    B2{"Two conditions"}
    C1["Human alone: diagnose complex vignettes\n(no AI)"]
    C2["Human + prompt-engineered LLM:\nstructured differential + rationale + next steps"]
    D1["Measures:\naccuracy/error type • time • confidence • trust/reliance • advice uptake"]
    D2["Analysis:\nwithin/between comparisons; subgroup by expertise & case difficulty"]
    E1["Contributions:\ncausal estimate of prompt-engineered support;\nhuman factors; reproducible prompt spec"]
  end

  Prior_Work --> Our_Study
  B1 --> B2
  B2 --> C1
  B2 --> C2
  C1 --> D1
  C2 --> D1
  D1 --> D2
  D2 --> E1
```

Links and suggested citation keys

Suggested author–year citations: **Goh2024**, **Zoller2025**, **Prinster2024**, **Patel2024**, **Wei2022**.

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Goh et al. (2024) JAMA Network Open (PMC): https://pmc.ncbi.nlm.nih.gov/articles/PMC11519755/
Zöller et al. (2025) PNAS (PMC): https://pmc.ncbi.nlm.nih.gov/articles/PMC12184336/
Prinster et al. (2024) Radiology (PMC): https://pmc.ncbi.nlm.nih.gov/articles/PMC11605106/
Patel et al. (2024) Scientific Reports: https://www.nature.com/articles/s41598-024-66933-x
Wei et al. (2022) NeurIPS (PDF): https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf
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1 5 7 10 11 16 17 22 Large Language Model Influence on Diagnostic Reasoning: A Randomized Clinical Trial - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11519755/

2 4 8 18 19 25 Evaluating prompt engineering on GPT-3.5's performance in USMLE-style medical calculations and clinical scenarios generated by GPT-4 | Scientific Reports
https://www.nature.com/articles/s41598-024-66933-x

3 9 15 24 26 Care to Explain? AI Explanation Types Differentially Impact Chest Radiograph Diagnostic Performance and Physician Trust in AI - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11605106/

6 12 13 20 23 Human–AI collectives most accurately diagnose clinical vignettes - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC12184336/

14 21 proceedings.neurips.cc
https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf